

J. FLETCHER *(they/them)*

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🏠 Scottsdale, Arizona

EDUCATION

Ph.D. in Mathematics

Arizona State University, Tempe, AZ

GPA of 4.00/4.00

Advisor: Jonathan Montaña

August 2024 - Expected May 2029

Passed Qualifying Exams: May 2025

Passed Comprehensive Exams: January 2026

B.S. in Mathematics

Arizona State University, Tempe, AZ

Barrett Honors College Graduate

Summa Cum Laude - GPA of 3.98/4.00

Phi Beta Kappa

Honors Thesis: *Cryptography in the Quantum Era: How the McEliece Cryptosystem May Be Used to Develop Quantum-Resistant Cryptography*

Honors Thesis Advisor: Florian Sprung

Certificate in Disability Studies

August 2020 - May 2024

RESEARCH INTERESTS

Interested in research in commutative algebra and its intersections with combinatorics and algebraic geometry.

RELEVANT COURSEWORK

Graduate Classes

MAT 512 Introduction to Combinatorics

Susanna Fishel

MAT 513 Graph Theory

Hal Kierstead

MAT 541 “p”-Adic Numbers

Nancy Childress

MAT 543 Abstract Algebra I

Jonathan Montaña

A graduate-level introduction to groups and rings.

3 credit hours.

MAT 544 Abstract Algebra II

Jonathan Montaña

A graduate-level introduction to modules, advanced linear algebra, fields, and Galois theory.

3 credit hours.

MAT 545 Abstract Algebra III (Introduction to algebraic geometry)

Jonathan Montaña

MAT 590 Reading and Conference (Commutative algebra)

Jonathan Montaña

Read about and worked through exercises on commutative algebra basics, such as chain conditions, localization and Spec, associated primes and primary decomposition, dimension and Hilbert’s Nullstellensatz, integral extensions, and introductory homological algebra.

2 semesters of 3 credit hours each.

TEACHING EXPERIENCE

Graduate Teaching Assistant

School of Mathematical and Statistical Sciences, Arizona State University, Tempe, AZ

Teaching

MAT 266: Calculus for Engineers II Expected January - May 2026

- Instructor on record for a Calculus II class with about 40 students.
- Responsible for preparing and administering lectures, quizzes, and group work.

MAT 271: Calculus w/ Analytic Geometry II August - December 2025

- Led one recitation with about 25 students.
- Responsible for making and facilitating weekly group work.
- Responsible for making, administering, and grading weekly quizzes and written homework assignments.

MAT 265: Calculus for Engineers I January - May 2025

- Instructor on record for a Calculus I class with about 40 students.
- Responsible for preparing and administering lectures, quizzes, and group work.

MAT 271: Calculus w/ Analytic Geometry II August - December 2024

- Led two recitations with about 30 students each.
- Responsible for making and facilitating weekly group work.
- Responsible for making, administering, and grading weekly quizzes and written homework assignments.

Grading

MAT 371: Advanced Calculus I August - December 2025

- Graded weekly homework assignments in Dr. Matthias Kawski's proof-based introductory real analysis course with about 30 students.

Lead Tutor

January 2023 - August 2024

Academic Support Network, Arizona State University, Tempe, AZ

- Developed review sessions for upcoming exams with practice problems designed to best improve student understanding.
- Led 1-hour review sessions over Zoom, going over practice problems and emphasizing key concepts.
- Moderated the online study hub Inscribe and answered questions students posted to better their understanding of the subject.

Subject Area Tutor

January 2022 - August 2024

Academic Support Network, Arizona State University, Tempe, AZ

- Assessed where a student may be struggling and gave them guidance to fill their knowledge gap, resulting in the student being able to succeed on their own in the future.
- Approached new problems with the student to facilitate their learning.
- Managed a queue of students both in-person and online to create greater tutoring accessibility and organization.

CONFERENCES ATTENDED

SLMath Graduate Summer School - Berkeley, CA, U.S.A.

July 2026

Singularities in Commutative Algebra through Cohomological Methods

- Attended a 2-week joint lecture series by Eloísa Grifo, Josh Pollitz, and Benjamin Briggs.
- Worked with other graduate students on exercises associated to the lecture.

Pan-American School in Commutative Algebra - Guanajuato, GTO, México

May 2026

- Attended lecture series' by Eloísa Grifo, Karl Schwede, Federico Castillo, and Daniel Duarte.
- Worked with other graduate students on exercises associated to the lectures.

SKILLS

- Intermediate Spanish Language (speaking, listening, reading, and writing).
- Typesetting in $\text{\LaTeX}$.
- Basic programming in Macauley2, Python, Java, Matlab, and Mathematica.

ADDITIONAL ACTIVITIES

- Cellist in Arizona State University's Philharmonia.